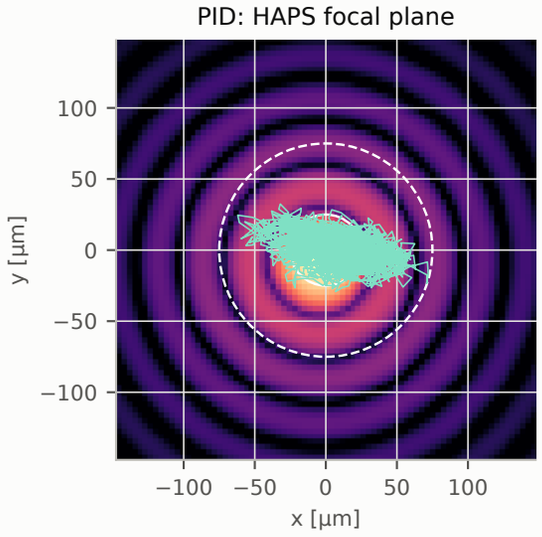
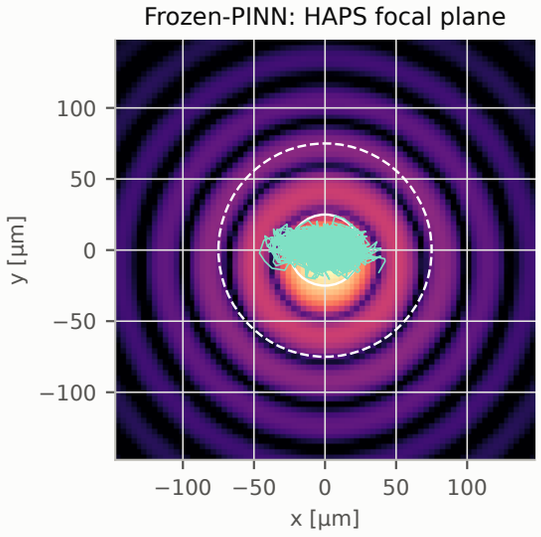
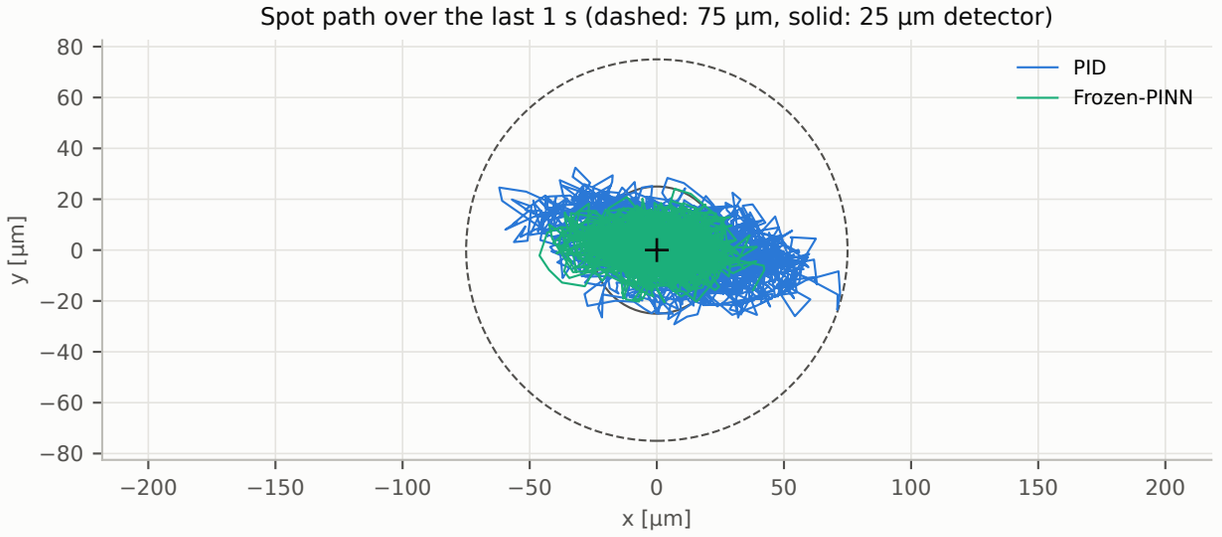
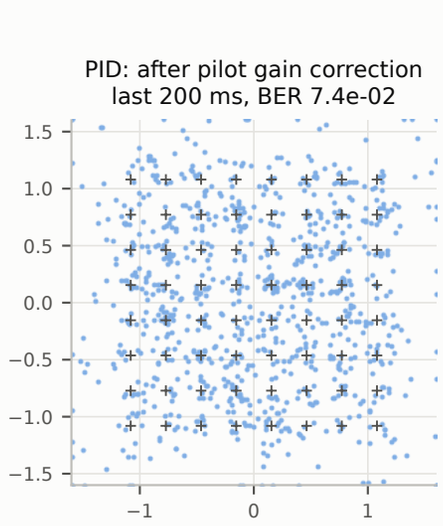
PAT + 64-QAM link | Injected 6-18 Hz vibration | t = 10.00 s

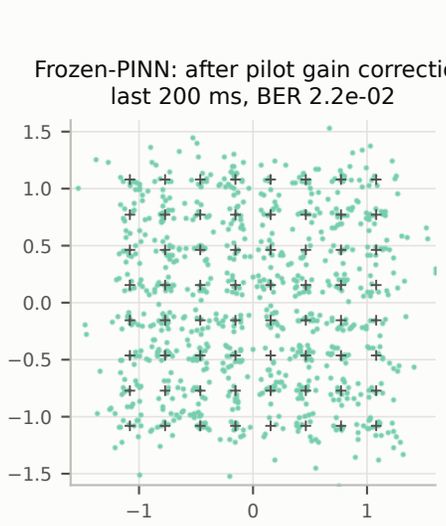
Spot: HAPS-receiver focal plane (log intensity) shifted by the residual. Symbols: 64-QAM at the operating SNR, pilot block every 0.5 ms.

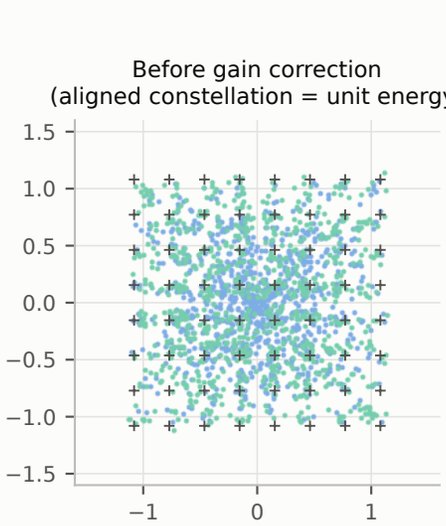













```
t = 10.00 s
Injected 6-18 Hz vibration
25 µm detector, aligned SNR
24.1 dB (FEC + 3 dB)

PID  (last 200 ms)
  RMS offset    22.5 µm
  mean power    0.603
  outage        61.6 %
  BER          7.4e-02

Frozen-PINN  (last 200 ms)
  RMS offset    14.0 µm
  mean power    0.803
  outage        24.4 %
  BER          2.2e-02
```

